

SMART ANDON SYSTEM FOR FAULT MONITORING AND DOWNTIME ANALYSIS

LEVEL 1 - PROBLEM ANALYSIS & PROPOSED SOLUTION

1.1 Problem Statement

Unexpected machine faults reduce production efficiency. Tool failures lead to machine stoppage and defective products. Material shortages interrupt continuous production. Motor overload may damage equipment and increase downtime. Sensor malfunctions reduce monitoring reliability. Manual fault reporting delays corrective actions. Downtime records are often maintained manually, making analysis difficult.

1.2 Industrial Evidence Supporting the Proposed System

Unplanned machine downtime is a major challenge in modern manufacturing industries, resulting in production losses, increased maintenance costs, and reduced equipment availability. Industries are increasingly adopting real-time monitoring and automated fault detection systems to minimize downtime and improve operational efficiency.

According to Siemens (2024), unplanned equipment downtime has a significant financial impact on manufacturing industries, highlighting the importance of intelligent monitoring systems and rapid fault detection.

Furthermore, Overall Equipment Effectiveness (OEE) is a widely accepted performance metric used to evaluate manufacturing productivity. Machine downtime directly reduces the Availability component of OEE, making continuous fault monitoring and downtime analysis essential for improving production efficiency.

The proposed Smart Andon System addresses these challenges by providing real-time fault detection, automated downtime tracking, fault occurrence monitoring and historical data analysis to support effective maintenance and production management.

References

- [1] Siemens AG, The True Cost of an Hour's Downtime: An Industry Analysis, 2024.
- [2] Vorne Industries, Overall Equipment Effectiveness (OEE).

1.3 Existing Solution

Manual monitoring by machine operators, verbal fault reporting, paper-based maintenance records, reactive maintenance after faults occur, and limited historical fault analysis.

1.4 Limitations of Existing Solution

Delayed fault detection, human errors in fault reporting, lack of real-time monitoring, no automated downtime tracking, difficulty in fault trend analysis and poor maintenance planning.

1.5 Proposed Solution

Develop a Smart Andon System for industrial fault monitoring capable of detecting faults in real time, classifying faults, automatically monitoring downtime, recording fault occurrences, maintaining historical fault records and generating analytical reports.

1.6 Faults Monitored

Tool Failure

Material Shortage

Motor Overload

Sensor Fault

1.7 Advantages of Proposed Solution

Real-time fault monitoring, faster fault identification, automatic downtime tracking, fault occurrence analysis, historical fault storage, improved maintenance planning, better machine protection and increased production reliability.

LEVEL 2 – ACTUAL INDUSTRIAL SMART ANDON SYSTEM DESIGN

2.1 Objectives

- Detect machine faults in real time.
- Classify faults as critical and non-critical.
- Protect industrial machines from failures.
- Monitor downtime automatically.
- Maintain historical fault records.
- Generate maintenance reports.
- Improve Overall Equipment Effectiveness (OEE).
- Reduce maintenance response time and production losses.

2.2 Industrial Block Diagram

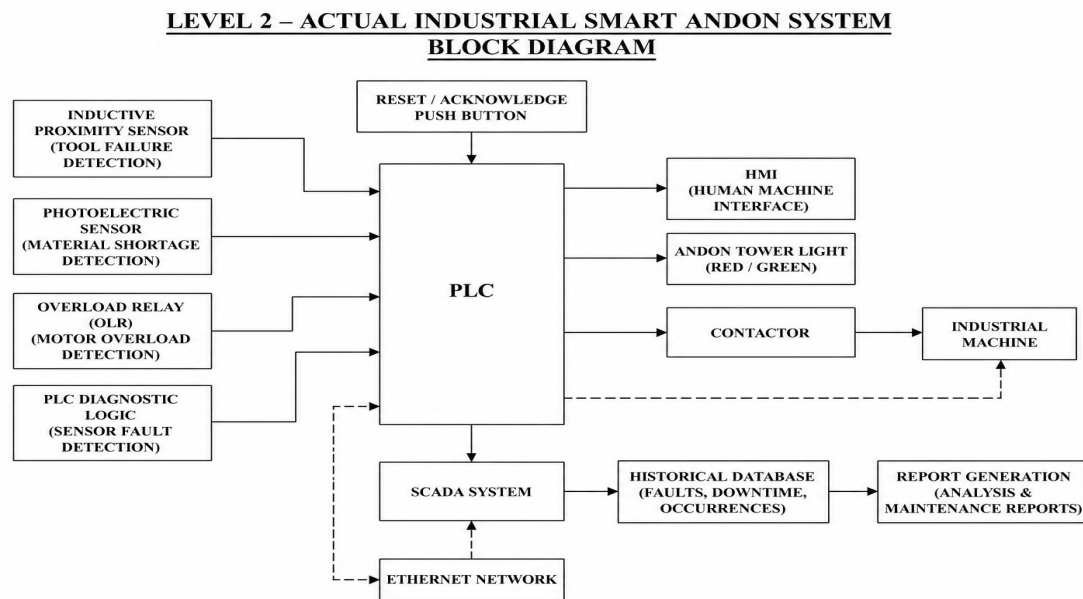


Figure 2.1: Actual Industrial Smart Andon System Block Diagram

The PLC receives signals from industrial sensors, classifies faults, activates Andon indications, protects the machine and communicates with the HMI and SCADA for monitoring and historical data storage.

2.3 Workflow Diagram

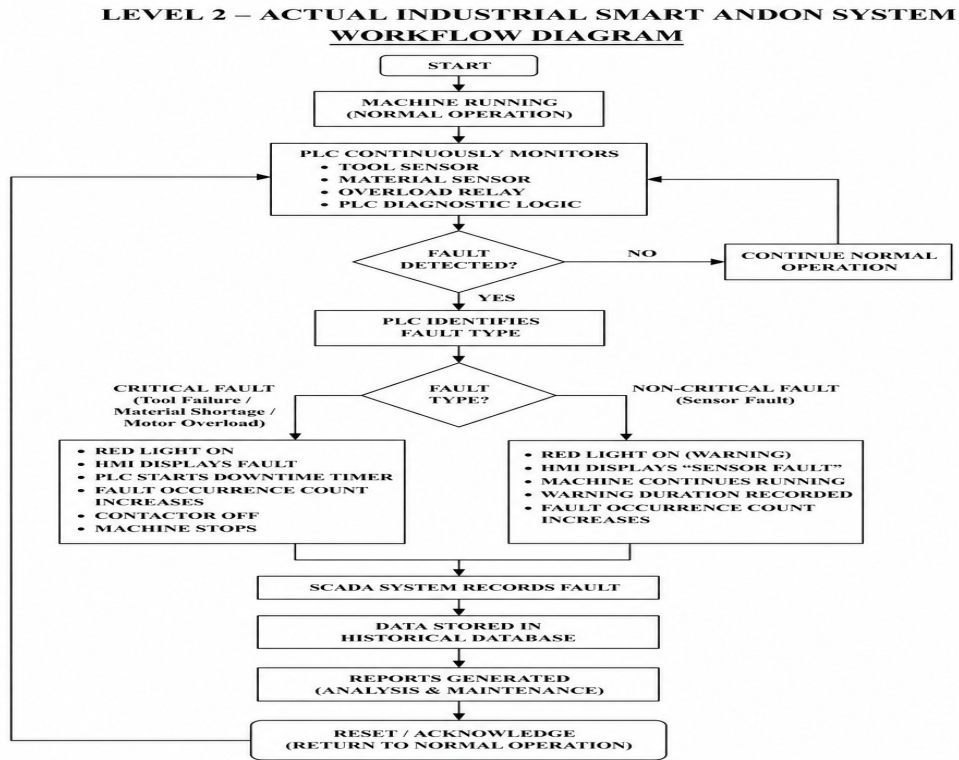


Figure 2.2: Workflow of the Proposed Industrial Smart Andon System

2.4 Components Used

Component	Function
Inductive Proximity Sensor	Tool Failure Detection
Photoelectric Sensor	Material Shortage Detection
Overload Relay	Motor Overload Detection
PLC Diagnostic Logic	Sensor Fault Detection
PLC	Central Controller
HMI	Machine Status Display
Andon Tower Light	Visual Indication
Contactor	Machine Protection
SCADA	Monitoring & Data Logging
Historical Database	Stores Fault Records

2.5 Fault Classification

Critical Faults: Tool Failure, Material Shortage and Motor Overload.

Non-Critical Fault: Sensor Fault.

2.6 Methodology

1. Industrial sensors continuously monitor machine conditions.
2. Tool Failure is detected using an inductive proximity sensor.
3. Material Shortage is detected using a photoelectric sensor.
4. Overload relay monitors motor current.
5. PLC diagnostic logic checks sensor health.
6. PLC receives all fault signals.
7. PLC identifies and classifies the fault.
8. Critical faults stop the machine and start downtime monitoring.
9. Sensor faults generate warnings while allowing operation.
10. HMI displays machine status and fault information.
11. SCADA records faults and downtime.
12. Historical database stores records.
13. Maintenance reports are generated automatically.

2.7 Expected Outputs

- Real-time fault detection
- Automatic fault classification
- Machine protection
- Automatic downtime monitoring
- Historical fault records
- Automated maintenance reports
- Improved maintenance planning
- Enhanced production reliability

PROOF OF CONCEPT (POC)

3.1 Objective

- Validate the proposed Smart Andon System using an ESP32-based prototype.
- Demonstrate real-time fault detection.
- Demonstrate machine protection during fault conditions.
- Demonstrate automatic downtime monitoring.
- Demonstrate automatic fault occurrence tracking.
- Demonstrate OLED-based machine status display.
- Demonstrate Serial Monitor data logging.
- Demonstrate Microsoft Excel-based report generation.
- Verify the feasibility of implementing the proposed industrial Smart Andon System.

3.2 Prototype Setup

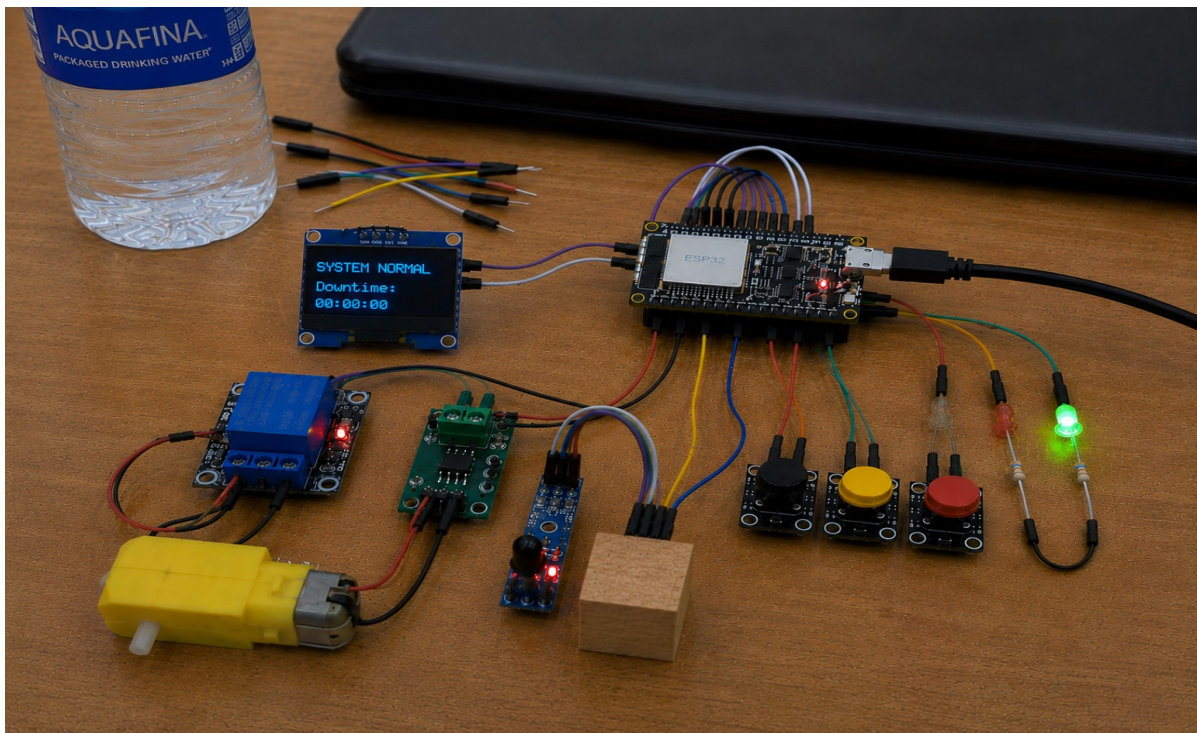


Figure 3.1: ESP32-Based Smart Andon Prototype

3.3 OLED Status Demonstration

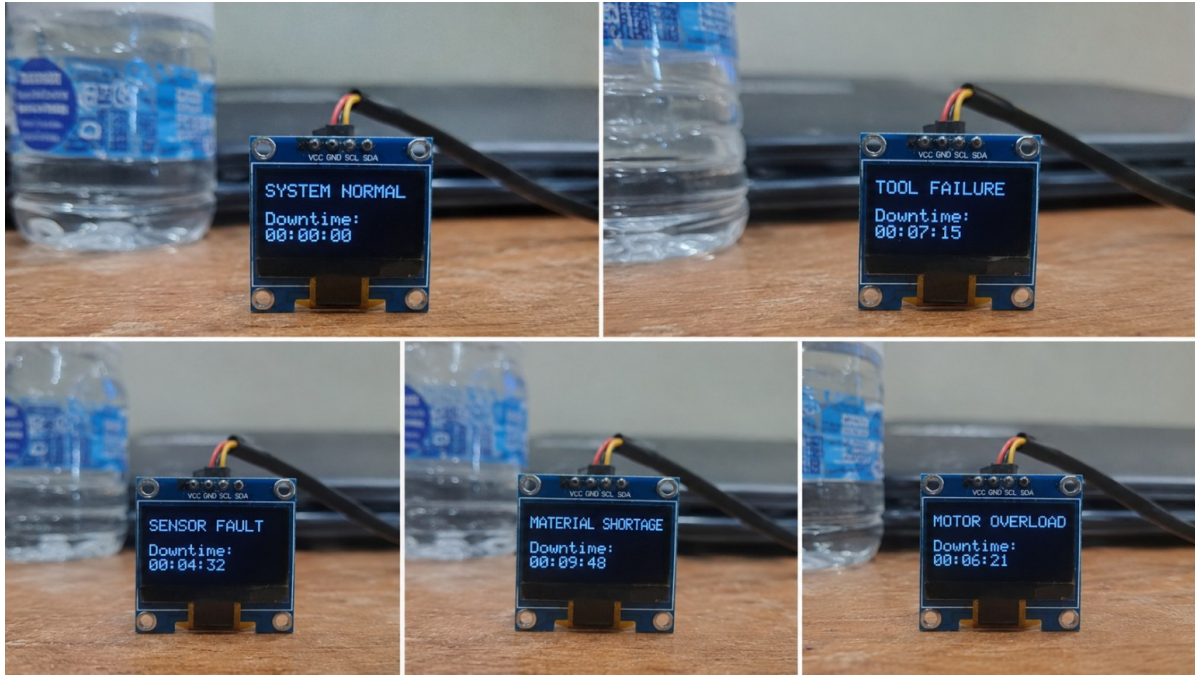


Figure 3.2: OLED Display Showing System Normal, Tool Failure, Sensor Fault, Material Shortage and Motor Overload Conditions

3.4 Components Used

ESP32, OLED Display, Tool Failure Push Button, Sensor Fault Push Button, IR Sensor, ACS712 Current Sensor, Reset Push Button, Relay Module, DC Motor, Red LED and Green LED.

3.5 Working Principle

1. The ESP32 continuously monitors all fault inputs.
2. Tool Failure is simulated using a push button.
3. Sensor Fault is simulated using a push button.
4. The IR sensor detects Material Shortage.
5. The ACS712 current sensor continuously monitors motor current.
6. When the measured current exceeds the predefined threshold, a Motor Overload fault is generated.
7. The OLED display continuously shows machine status, detected fault and downtime.
8. During Tool Failure, Material Shortage, Sensor Fault and Motor Overload conditions, the relay module disconnects the DC motor to demonstrate machine stoppage.

9. Automatic downtime monitoring starts immediately after a fault is detected.
10. Fault occurrence count is updated automatically.
11. Fault information is transmitted through the Serial Monitor.
12. The collected fault data is transferred to Microsoft Excel for report generation.
13. After the fault is cleared, pressing the Reset button restores the system to normal operation and resumes continuous monitoring.

3.6 Data Logging and Report Generation

Fault information is transmitted through the Serial Monitor, recorded, transferred to Microsoft Excel and used to generate weekly fault analysis reports.

3.7 Results and Analysis

	A	B	C	D	E	F	G	H	I	J
1	Day	Tool Failure Occ.	Tool Failure DT (min)	Sensor Fault Occ.	Sensor Fault DT (min)	Material Shortage Occ.	Material Shortage DT (min)	Motor Overload Occ.	Motor Overload DT (min)	Total DT (min)
3	Day 1	1	8	0	0	0	0	0	0	8
4	Day 2	0	0	1	6	0	0	1	7	13
5	Day 3	0	0	0	0	1	10	0	0	10
6	Day 4	1	7	1	5	0	0	0	0	12
7	Day 5	0	0	1	8	1	6	1	9	23
8	Day 6	2	12	0	0	1	5	1	8	25
9	Day 7	1	6	2	9	1	4	0	0	19
10	Total	5	33	6	28	4	25	3	24	110
11										
12										
13										
14										
15										
16										

Figure 3.3: Weekly Fault Analysis Report

Fault	Occurrences	Downtime (min)
Tool Failure	5	33
Sensor Fault	6	28
Material Shortage	4	25
Motor Overload	3	24
Total System Downtime	18	110

3.8 Observations

The prototype successfully detected all four fault conditions, stopped the motor during critical faults, displayed machine status on the OLED, automatically tracked downtime and generated Excel-based fault reports.

3.9 Note on Accelerated Testing

The proposed Smart Andon System is intended for continuous industrial operation where machine faults occur over long production periods. Since long-duration testing is impractical in a laboratory, an accelerated testing methodology was adopted. One second of prototype testing was considered equivalent to one minute of industrial machine downtime. This enabled multiple fault scenarios to be validated while preserving the actual fault detection, machine protection, downtime monitoring and report generation logic.

3.10 POC Outcome

The prototype successfully demonstrated Tool Failure detection, Sensor Fault detection, Material Shortage detection, Motor Overload detection, machine protection, automatic downtime monitoring, fault occurrence tracking and Excel-based report generation.